Intervals and absolute value equations

Two-vertex certification of nonsingular sign regularity

Difficulty: challenging **Importance:** interesting to specialist

Status: open; checked 2026-09-08
Area: structured matrices and interval linear algebra

Problem statement

For $\epsilon \in \{-1, 1\}^n$, call a real $n \times n$ matrix M **nonsingular sign regular with signature** ϵ if $\det M \neq 0$ and every minor of order k has determinant d satisfying $\epsilon_k d \geq 0$, for every $1 \leq k \leq n$.

Let $n \geq 5$ and let $L, U \in \mathbb{R}^{n \times n}$ satisfy $L \leq U$ entrywise. Define the two checkerboard vertex matrices

$$A^-_{ij} = \begin{cases} L_{ij}, & i+j \text{ even}, \\ U_{ij}, & i+j \text{ odd}, \end{cases} \qquad A^+_{ij} = \begin{cases} U_{ij}, & i+j \text{ even}, \\ L_{ij}, & i+j \text{ odd}. \end{cases}$$

Question

If A^- and A^+ are nonsingular sign regular with the same signature ϵ, must every real matrix M with $L \leq M \leq U$ be nonsingular sign regular with signature ϵ?

The implication would certify a property of an entire interval of matrices from only two of its vertices. Allowing zero minors is essential to the remaining problem.

References

Jürgen Garloff, Mohammad Adm, and Jihad Titi, *A Survey of Classes of Matrices Possessing the Interval Property and Related Properties*, Reliable Computing **22** (2016), 1–14, Conjecture 3.1, p. 7. Adm and Garloff, *Certification of the Sign Regularity of Matrix Intervals*, Acta Scientiarum Mathematicarum, published January 17, 2026, §3, especially the question following Theorem 3.2 and the discussion of admissible signatures.

Status evidence

The 2026 survey explicitly retains this question. It records positive results for strictly sign regular matrices, totally nonnegative matrices, tridiagonal matrices, and certain signatures covering all orders at most four. Theorem 3.3 also covers intervals whose fixed entries $L_{ij} = U_{ij}$, if any, all have the same parity of $i + j$; fixed entries of both parities are allowed in the general question. It also records counterexamples when nonsingularity is omitted. Searches for sign-regular interval conjectures and subsequent 2026 results found no resolution of the displayed implication.